Trio laser hair removal machine

Welcome to use this laser beauty hair removal machine. This revolutionary diode laser beauty hair removal device provides a smooth, comfortable, long-lasting effective hair removal, skin rejuvenation, whitening, and red bloodshot experience for a variety of skin types.

Contents

Welcome page 0

1. Company profile..... 1

2. Notice..... 2

3.Packing list 3

4.Technical Parameters.....4

5. System structure and circuit structure diagram..... 5

6. Instructions..... 6

7. Interpretation of the interface.....7

8. Equipment maintenance.....8

9. Quality assurance clause.....9

10. Simple fault solutions.....10

11. Attachment 11

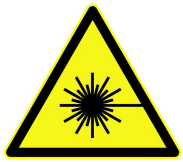
1. Company Profile

Founded in 2008, Oriental-laser (Beijing) Technology Co., Ltd. is a high-tech enterprise specializing in the R&D, production and sales of laser equipment. The headquarters and R&D

center are located in Daxing District, Beijing, the “China Silicon Valley” of science and technology and political centers, and have branches in Hong Kong. Over the years, it has absorbed high-end talents in various fields of DPSS, focusing on high-power semiconductor side pump module (DPSS Laser), high-power semiconductor laser products, semiconductor laser cosmetic equipment development and production, and system integration, its products in industrial, cosmetic, It has a wide range of applications in research and military fields.

The company has 500 square meters of overall 10K class, local 1000 class of ultra-clean workshop, strictly meets the US 109E federal industrial standards and obtained ISO9001 quality management system certification. After years of team research and experience, Oriental-laser has DPSS semiconductor pumped solid-state laser design, high-power semiconductor vertical/horizontal/profiled array package laser, semiconductor laser cosmetic equipment system integration, DPSSL semiconductor pumped solid-state laser design and integration. Production lines such as application systems.

Based on independent research and development, the company is determined to become a professional and strong core competition based on the enterprise spirit of “integrity, professionalism, dedication and innovation”, scientific management, leading technology, standardized production, excellent service and good reputation. The company.
The company attaches great importance to product innovation, process control and quality management. By using our products or solutions, Oriental-laser's customers have won rich profits and praise. For this reason, Oriental-laser (Beijing) Technology Co., Ltd. has accumulated a stable customer base worldwide for many years. In the future, we will continue to adhere to the goal of “trying to make a profit for our customers” and strive to provide quality products and services to more enterprises.



2. Precautions

1. After receiving the machine for 24 hours, it can be turned on: check the accessories, assemble the machine, add water to the machine, and let it stand for 24 hours. (If you are not starting up in advance according to the regulations, please solve the problem yourself.)

2. The machine needs to add pure water, and it needs to be replaced once a month. It needs 2.5~3 bottles. If the instrument needs to be idle for more than half a month, it needs to be drained and then idle.
3. Please read this operation procedure carefully before receiving the machine, carefully study the operation video, precautions and other information, then turn on the operation.
4. Before the instrument moves, it needs to be drained. It cannot be tilted and cannot be inverted. After 24 hours of moving, the machine is turned on, the operation is not performed according to the requirements, the machine is damaged, and the problem is solved by itself. (Long-distance mobile equipment, you need to notify the corresponding sales staff before you can move.)
5. After the machine is received, there is a red button at the bottom of the screen. Please twist in the direction of the arrow on the red button until the red button pops up.
6. Insert the key and turn on the machine.
7. The patient must completely cover the eyes with an eye mask, and therapist must wear laser safety glasses.
8. After the skin is cooled, it takes about a minute to wait for the treatment head to cool.
9. For non-treatment, place the hand on the bracket on one side of the main unit and keep the treatment head down. In order to prevent the condensation water at the front end of the handpiece from flowing back to the inside of the handpiece, the laser is damaged.
10. At the beginning and end of treatment, please pay attention to clean the treatment head. Never leave cold gel, hair, or dander on it, it will burn the laser head.
11. The equipment environment should be equipped with air conditioning, and the room temperature should be below 28 °C.
The ambient temperature when the equipment is idle should be greater than 0 °C. If the water circulation system inside the equipment freezes, it will directly cause equipment damage.

3. Packing list

product name	Quantity	Remarks
Beauty equipment host	1 set	
Beauty handle	1 piece	
Foot switch	1 piece	

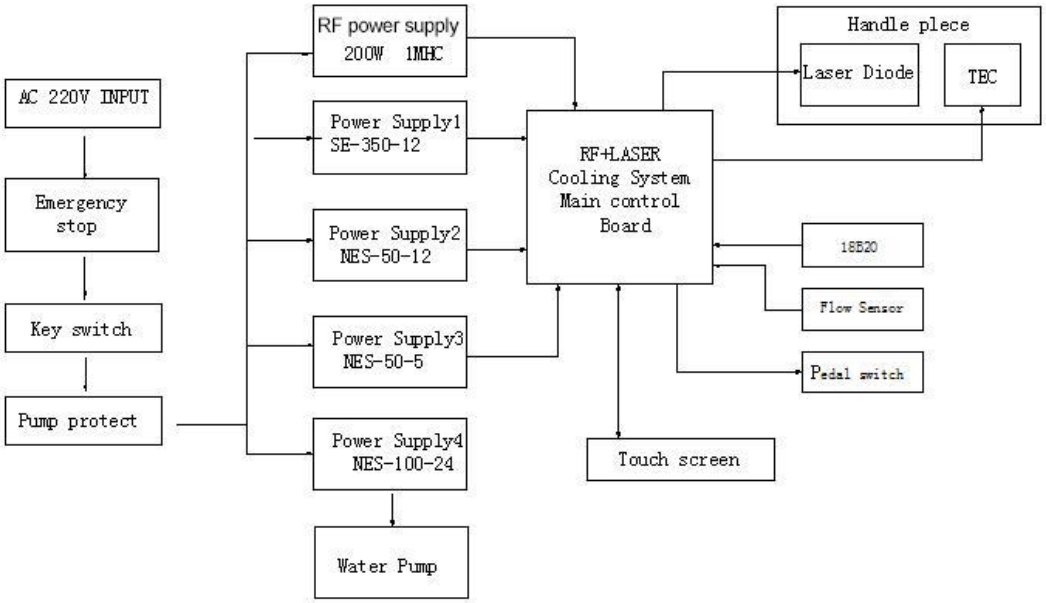
VAC line (power cord)	1 item	
Water pipe	1pc	for water injection and discharge
Funnel	1pc	
Key	2 pcs	
Packing List	1serving	

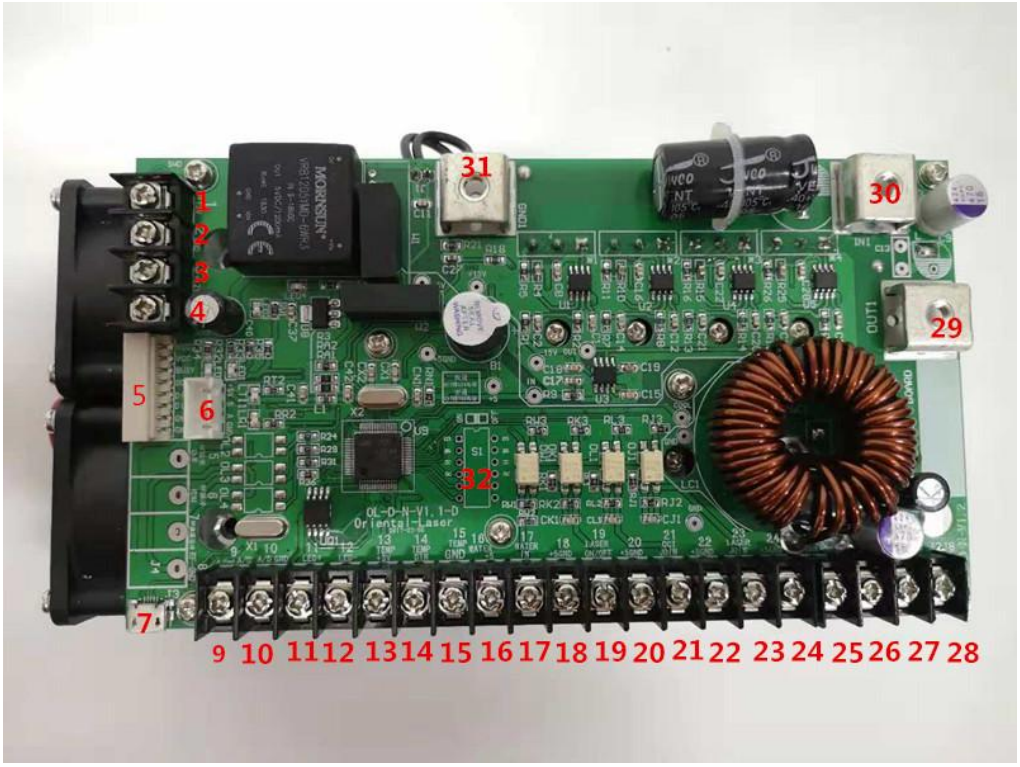
4.Technical parameters

3.1 Electrical parameters

Model	OL-HR-ROSE
Energy	0~120J/cm2
Wavelength	808nm/755nm/1064nm
Pulse width range	0ms~600ms
Frequency Range	1Hz~10Hz
Sapphire operating head temperature	-5℃--5℃ by cooled Sapphire
Laser head size	15x25 mm
Interface language	ENGLISH
LCD screen size	10.4’
System cooling method	R134A Freon compressor refrigeration
Power	220/110VAC / 50-60Hz current 10A
Cooling water requirement	Deionized water
Ambient temperature	15~28℃
Environment humidity	<70%
Packing Box Dimension	123(L)x64(W)x64(H)cm
Volume Weight	65KG

5. Equipment structure and circuit connection





Number	Description
1,2	1 and 2 are the power input ports for the control board. The supported voltage range

	is 9-18V. 1 is positive and 2 is negative.
3,4	Pins 3 and 4 are 12V outputs. The default fan is attached.
5	The 5 feet are the display access ports.
6	The 6-pin is the inlet for counting the small board.
7	7 feet is the download port of the update program.
9,10	The 9 and 10 feet support the 4-20ma current signal to collect the voltage.
11,12	The 11 and 12 feet can control the LED lights.
13,14,15	Pins 13, 14, and 15 are the access ports for the digital temperature sensor (DS18B20).
16,17,18	The 16th, 17th and 18th feet are the access ports of the Hall flow sensor.
19,20	Pins 19 and 20 are the access ports for hand switches or pedals.
21,22,23,24	21 and 22 or 23 and 24, the input switch signals reserved for the two paths.
25,26,27,28	The 25, 26, 27 and 28 feet are the control feet of the cooling TEC. Among them, 27, 28 are the power input ports of the TEC, 27 is positive and 28 is negative. 25, 26 to TEC. The TEC control module of this control board can be adapted to most models of TEC. And the cooling capacity is adjustable.
29,30,31	The 29, 30 and 31 pins are the access ports for the laser and laser power supplies. Among them, the negative electrode of the 29 laser, 30 is connected to the positive electrode of the laser and the laser power supply, and 31 is connected to the negative electrode of the laser power supply. This control board supports up to 100A of output.
32	32 is a reserved dial switch.Size 200*100*95

6. Instructions

Please read this carefully before you start the device for the first time.

6.1 Standing before using

After the instrument receives or moves, it must be standing for 24 hours to start. If it is not turned on according to the requirements, take responsibility for yourself.

6.1.1 Standing after receiving

Because large instruments have compressors, during long-distance transportation, bumps and tilts will inevitably occur, causing the lubricating oil in the compressor to flow out. It is necessary to let the lubricating oil flow back for 24 hours before using it. Lubricating oil can protect the compressor. If it flows into the refrigeration system, it will seriously affect the cooling effect.

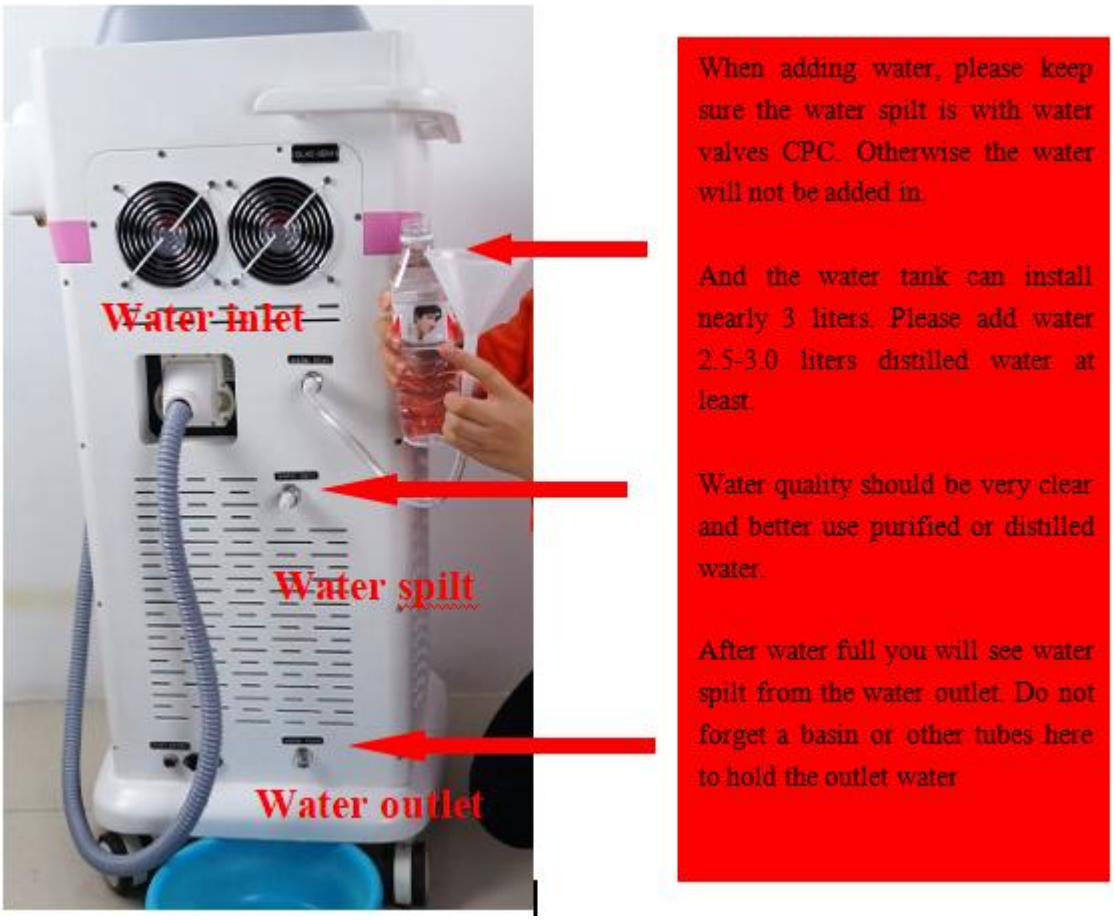
6.1.2 Standing before moving

Please do not tilt or move long distances. Do not tilt or invert when the instrument moves or needs to be drained. If it is tilted or moved, it needs to be left for 24 hours before it can be used.

6.2 Instrument assembly

6.2.1 Water injection operation

- Oriental laser workstation needs to change the cooling water every month (use pure water, no use of mineral water or tap water), Wahaha pure water 500ml can be loaded with 3 bottles.
- Overflow: Insert a short spout into the overflow and insert it successfully after hearing a click. The nozzle is inserted into the overflow to balance the pressure in the equipment tank.
- Injecting the nozzle: Keep the nozzle that was originally inserted in the overflow port, and use the thin end of the water pipe with the funnel to insert it into the water inlet. After hearing the click, the insertion is successful.
- Full water indication: use the funnel to slowly inject pure water into the equipment until there is water in the overflow, and the water in the tank is full.
-



-

6.2.2 The water discharge operation is as follows:

- Connect the drain: Insert a short spout into the drain, and insert it successfully after hearing a click.

- Drainage operation: After inserting the overflow water pipe, it will automatically discharge water. Be sure to use the basin to catch the waste water.
- Release the instructions: until there is no more water flowing out of the drain, the water in the tank has been drained.

6.2.3 Mounting handle bracket

The three sides of the housing on the side of the device are marked with three screw holes. Use the countersunk head screws to install the handle bracket.

6.2.4 Connection handle

Use your fingers to press the two dots outside the interface of the handpiece and press the product firmly into the interface until you hear a “click” sound from the quick-injection connector and confirm that there is no water leak. Note: When you need to disassemble the laser hand, use your fingers to press the two dots on the interface and pull the product out. Be sure to repeatedly confirm that the handle is firmly connected, there is no water leakage, and the connection is not strong, which may cause serious consequences such as equipment alarm and pump damage.

6.2.5 Power interface

Connect one end of the power cord to the instrument main unit and the other end to the power strip. Note: For the first time, tearing off the protective film on the screen will help you operate the screen more sensitively.

6.3 Preparation for hair removal

6.3.1 Keep a photo:

For all hair removal or beauty customers, before the first use (or before shaving), please take photos in bright light for comparison of post-effects.

6.3.2 Part shaving:

- Require customers not to have sun exposure within a week, and no other skin damage;
- If there is makeup or skin care in the hair removal area, you need to remove makeup or clean it

first;

- Instruct customers to gently remove the hair from the hair removal area, pay attention to gentle use, and do not scratch the surface skin; do not use hair removal to remove hair, such as: wax removal, hair removal, etc.; can not use hair removal cream to remove hair.

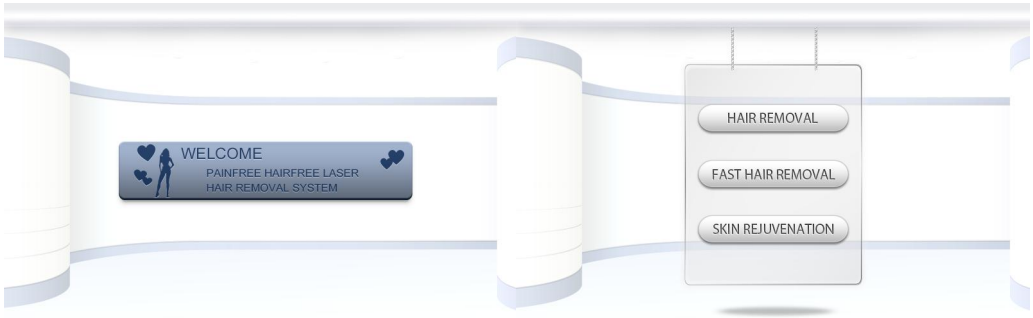
6.3.3 Skin cooling:

Apply a thick layer of cold gel to the customer's hair removal area. If you need to treat sensitive areas, in order to further reduce the pain, you can apply ice before treatment.

6.4 Laser hair removal project operation

6.4.1Starting up

(1) Turn the key to the right to turn it on. In case of emergency, you can immediately press the red emergency stop button. If you press the emergency stop before, the emergency stop must be turned to the right for half a turn to reset the bounce, in order to restart.
(2) During operation, you can quickly turn off the device by pressing the emergency stop button on the device.
(3) For the bilingual interface machine, select Chinese, click the welcome interface, Chinese interface machine, and click the welcome interface directly.
(4) When you need to turn off the device, turn off the key switch. To ensure safety, you can also unplug the power cord.



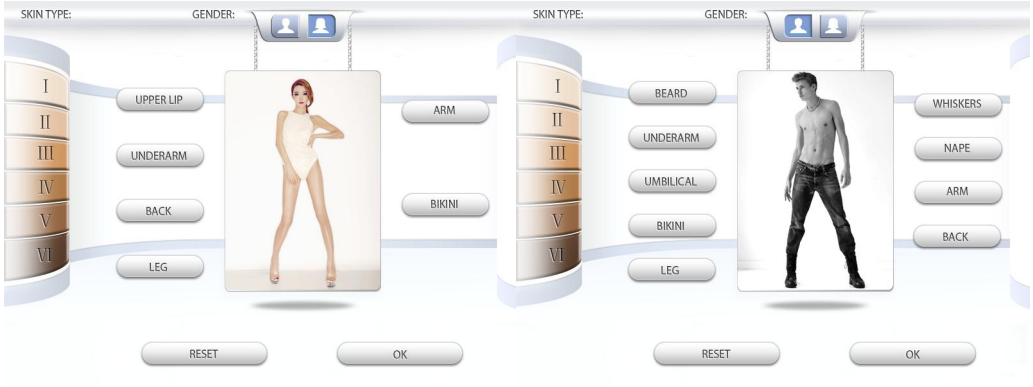
Interface 1 Interface 2

6.4.2 Select item:

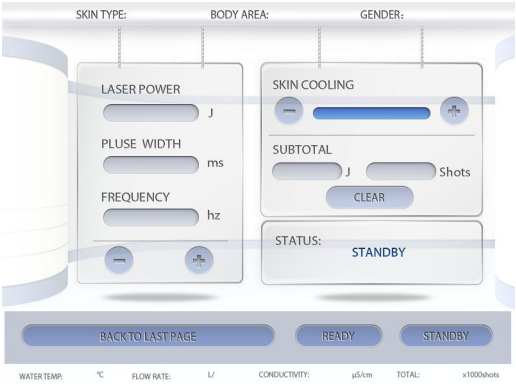
You can choose from three operating modes. They are HR (Hair Removal), FHR (Fast Hair Removal), and SR (Skin Rejuvenation). By selecting different modes, you can enter the corresponding interface. First select the "Professional Hair Removal" project highlights, other projects are more simple, we will introduce later.

6.4.3 Select function:

(1) gender, (2) skin color, (3) hair removal site,
According to the customer's actual situation, after making a choice, click Confirm.



Interface 1-2 interface1-3



6.4.4 Parameter Setting

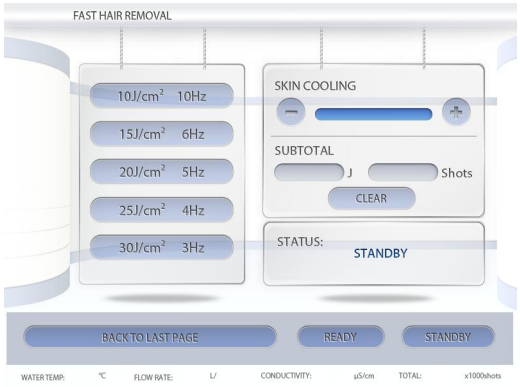
According to the selection of different parts, the starting value of energy is different:
(1) for dark hair, dense hair and sensitive parts, the energy must start from a low number, such as 6J, and the frequency must start from 1.
(2) for light hair, sparse hair and insensitive parts, the energy can start from a general number, such as 10J, and the frequency can start from 2 or 3.
(3) the pulse width cannot be set and does not need to be set. As long as the energy and frequency are selected, the pulse width will be automatically adjusted.
(4) the skin cooling parameters must be increased to about 80% to 90%.

6.4.5 Operation Treatment

When the setting is completed, click "Ready". The preparation time is 30 seconds at most. If the setting time is more than 30 seconds, the user will enter the stand by state directly.
(1) constantly adjust the energy and frequency when the customer is preparing for hair removal. It is recommended to adopt a higher energy density that the customer can tolerate and slide treatment. Be sure to keep the handle head in full contact with the skin to allow it to cool better and reduce

pain. During operation, the laser pulse can be gently regenerated to achieve the best hair removal effect. When depilating the client has prickly feeling or by ant light the feeling that bite is optimal

It can make the laser pulse overlap slightly to achieve the best hair removal effect. It is best for the customer to feel tingling or to be bitten by ants when hair is removed.
(2) record the adjusted laser parameter state, and the customer can use the parameter directly for hair removal at this part next time.
(3) The higher the energy and frequency, the better the effect. Generally, the large tingling that the customer can endure is appropriate, and should not be too large to prevent burns to the customer.
(4) the general parts are sliding operation depilation, in addition to the individual position is too small or not suitable for sliding, can be fixed point operation.
(5) the specific operation time or total energy of each part is written in the parameter Settings in the next two pages for your reference.



6.4.6 Fast hair removal：

(1) Only need to adjust the skin cooling, other parameters have been set. Skin cooling parameters must be increased to about 80% to 90 %
(2) For any body part, be sure to start from the lowest energy level.

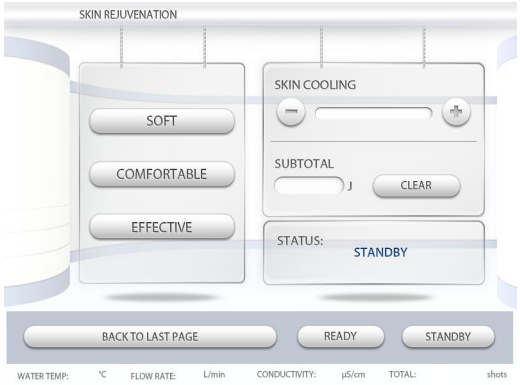
6.4.8 Laser Hair Removal Principle

(1) laser hair removal USES melanin to absorb laser light energy, produce thermal effect and destroy hair follicles.Because the process of hair follicle necrosis is irreversible, laser hair removal can achieve permanent hair removal.
(2) hair has a growth cycle: growth - degradation - resting period, and then the next cycle. Hair follicle that is in partly degenerate period and resting period, do not contain melanin, laser cannot destroy it. The natural growth cycle of hair on average, is 6 to 8 weeks. Therefore, it is recommended to extend the interval cycle of hair removal to 21-30 days, after 5-8 times of treatment, lasting for more than half a year, so that the hair follicles in the stage of degradation and rest cycle to the growth period, all destroyed, you can achieve permanent hair removal. (if there are still some tiny hair after half a year, you can let the customer consolidate treatment 2 to 3 times to achieve complete permanent hair removal

6.4.9 repair treatment

(1) after hair removal, you can use ice compress or aloe vera gel to soothe your skin.
(2) do not change the daily skin care products after hair removal, and try to avoid irritating cosmetics and color cosmetics.
(3)Skin has prosthesis or do uric acid and other micro-needle injection, it is not recommended to use.
(4)After depilatory very few client appear rash, belong to normal phenomenon, wait for 3-7 days to disappear naturally.
(5)The place after depilate, cannot hot water is stimulated inside a day, cannot insolate inside a week, summer needs to use sunscreen, and unfavorable go out for long.
(6) do not drink alcohol and avoid seafood, spicy and other stimulating foods
(7) for customers who have used hair removal methods such as honey wax, laser hair removal can only be used after the hair is completely growth after 6 months.
(8) hair may be easy pulled out after 1-2 times of laser hair removal, but the hair cannot be pulled out. The laser cannot conduct heat to the hair follicle without hair, which will directly affect the hair removal effect.

6.5 Laser skin rejuvenation project operation



6.5.1 Laser skin rejuvenation operation method

(1) Just adjust skin cooling, skin cooling parameters must be added to 80% to 90%.
(2) All other parameters have been set. New customers start with "soft whitening."

6.5.2 Laser Beauty Project Operation - Principle

The laser main application wavelength of Oriental-Laser is 808-810nm. The color base that absorb such near-infrared light include melanin, hemoglobin, and water. The laser produces a thermal effect by targeting the absorption by these three color bases, resulting in the following skin rejuvenation effects:

（1）The semiconductor laser acts on collagen cells and Connective tissue, in the dermal tissue to induce the synthesis of new collagen cells and extracellular matrices, and to improve wrinkles,

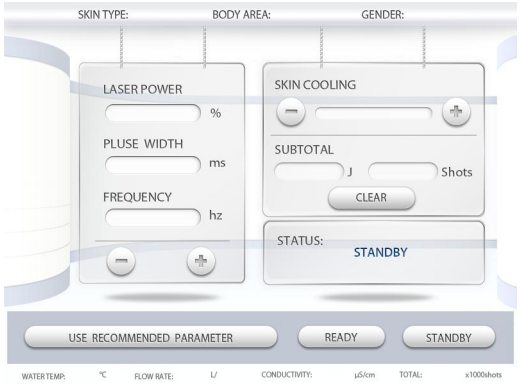
pit scars, and elastic tissue degeneration without disturbing or damaging the epidermis.

（2）While promoting the structural changes of dermal collagen and contractile tissue, the semiconductor laser also selects melanin, capillariesin the dermal tissue, to cause thermal decomposition. Reduces pigmentation of the dermis and epidermis, reduces telangiectasia, and improves smoothness and gloss throughout the skin. Problems related to improvement include: photoaging, senile plaques, red blood, dull skin, diffuse erythema, and rosacea and other related vascular and pigmentary problems.

6.5.3 Laser Beauty Project Operation - Follow-up Treatment

(1) Each full face treatment time is 10 to 20 minutes, as for 10 minutes or 20 minutes, depending on the customer, if the condition is light, it is 10 minutes, if the situation is more serious, it will be 20 minutes. The two treatment intervals are generally 15 days. We have time interval suggestions on the next page.
(2) Treatment outcomes: All aspects of photoaging including telangiectasia, irregular pigmentation, wrinkles, rough skin and enlarged pores showed significant improvement in more than 90% of patients.
(3) There is no downtime during treatment, no plaques appear, and a small number of patients have side effects such as edema and transient pigmentation changes, which can be recovered by themselves.
(4) In the skin after laser beauty, new collagen synthesis can be found.

7. Definition of the screen interface parameters



7.1 Fluence：shows the energy value of a single pulse laser per unit area. The laser energy density can be adjusted between 1-120J.

7.2 Frequency: shows the pulses per second of the laser. The frequency can be variable from 1Hz~10Hz with a step width of 1Hz. In the case of a selected energy density, the system

automatically defaults to the maximum frequency achievable at that energy density. You can also adjust the frequency by the "+" and "-" keys.

7.3 Subtotal：at the top-right of the LCD screen shows the total laser energy of this treatment.It is easy for you to calculate how much power used in one treatment.Press button “clear” can clear the number. Or the subtotal number will clear automatically when machine shut down. Through this function, you can refer to the energy values of the different parts of the treatment in Table 6.4.7 for comparison and measurement.

7.4 Skin Cooling Level shows cooling efficiency of the Sapphire Window and Aluminum Tip of the handle piece. OL-DHR-MI-808uses high quality sapphire contact refrigeration technology. When operating, the hand head is in contact with the skin, and the skin temperature is controlled at 0-5 ° C, which reduces the pain of the epidermis and truly obtains a painless operation process.The default cooling level is minimum output, which can be adjust by pressing “+” and “-” button.The green block is longer, the cooling is better. Skin cooling is turned off by default when the machine is turned on.

7.5 Status, shows the current device working condition，including：

7.5.1 Status: standby, In the “standby” state, the operator can adjust the skin cooling level at any time according to the patient. However, when in the "Ready" or "Working" state, the cooling level can not be adjusted. You may take 1-2 minutes to wait for the laser head to cool.

7.5.2 Status: ReadyReady for treatment. When all the parameters are set and you are ready to operate, press the “READY” button, the system will enter ‘**Ready**’ status. In this status, the laser is ready to use. At this time, press the button on the handpiece or press the foot pedal to operate.

7.5.3 Status: Working: When the button on the handpiece is pressed or the foot switch is pressed, the system enters the working state.

7.5.4 The status bar also displays some alarm indications when the entire beauty instrument is malfunctioning. Device alarm troubleshooting will be explained in detail in the next section.

7.6 Status at the lower of the screen:

7.6.1 Water temperature refers to the temperature of the cooling water inside the current equipment.

7.6.2 Flow Refers to the flow rate of cooling water inside the equipment.

7.6.3 Ionicity: Refers to the conductivity and cleanliness of the cooling water inside the equipment.

7.6.4 Total Shots: refers to accumulative shots numbers of the current treatment handle.

8. Maintain

8.1 This equipment requires pure water for cooling. When filling the equipment, please add pure water strictly. Otherwise the device will be damaged soon.

8.2 The machine must be operated in a room with air-conditioner, and the room temperature should be between 5 ° C and 28 ° C.

1. The environment temperature can not go below ZERO, or the ice may damage the water tanks and water channels inside the machine and the handle piece.
2. Clear the gel on the handle piece and make sure there is no dust around the Sapphire.
3. Do not bend the tube of the handle piece too hard, or this may cause water flow rate low warning.
4. Do not plug in & out the handle piece connectors frequently. This may cause water leakage or water jam or bad electrical connections.
5. Do not shot the laser at random or it may cause burning or fire.
6. Always keep the handle piece with the holder on the main machine when you don't use it. Otherwise, the condensing water may flow back to the laser stack in handle piece, so that cause laser failed.

9. Warranty notes

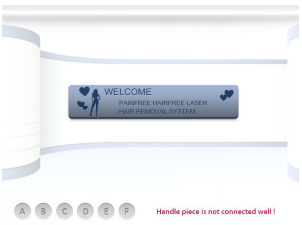
Please operate the equipment strictly in accordance with the above precautions and notice that following situations can **not be warranted:**

1. Operation the system without water.
2. Power up the system without connecting the handle piece.
3. Regular drops of the handle piece.
4. Wrong input VAC.
5. Other manual operating mistakes.

10. Fast troubleshooting and solution

After the device is turned on, the system will run the self-test program at all times. If there are abnormalities in the various components of the system, the alarm will be displayed:

10.1 The alarm prompt that will appear on the boot welcome screen:

Prompt picture	Interpretation	solution
	laser handpiece hasn't been inserted well	1. The laser handpiece hasn't been inserted well, and the contact is bad. 2. The small circuit board inside the plug is damaged, contact the manufacturer for replacement. Update the program to mask the circuit alarm settings in the handle. (emergency means, not recommended)

10.2 Enter the alarm prompt of the setting interface:

Prompt picture	Interpretation	solution
	Status: Water temperature high. Over temperature alarm 。The cooling water temperature range should be between 15~28℃. If it higher than 34℃, the laser will stop.	1. Observe the reading of the machine's back panel thermometer if the reading does exceed 34 °C. And if the compressor is normally turned on, it may be because when just turn on the machine, the compressor just start working, the ambient temperature is too high , resulting in too high water temperature. At this time, it is necessary to reduce the room temperature to within 30 degrees Celsius, and wait for about half an hour, the refrigeration system will gradually decrease the water temperature low than 34 °C. 2. Observe the reading of the machine's back panel thermometer: Confirm if the compressor is starting. Touch the compressor with your hand to see if there is vibration, or listen to the sound to determine if the compressor is starting. If the compressor does not start properly, contact the factory for assistance. 3.Confirm that if the tank has enough water. If the water injection is insufficient, there will be a relay in the compressor circuit that frequently trips, indicating that the water in the tank is insufficient.If there is enough water, the compressor starts up and works normally, and the temperature on the back panel is also less than 34 degrees, it may be that the temperature probe connected to the motherboard is damaged, or the temperature probe is virtual connection to the motherboard connection line,

		or The main control board is damaged. The solution is as follows: 1) Check if there is a virtual connection between the probe and the circuit. 2)Replace the board. Please refer to the attachment or click this link for the board replacement step. 3)If the replacement of the board still does not solve the problem, it is determined that the probe is damaged and the probe needs to be replaced. To replace the probe, you need to open the entire machine and open the water tank.**Refer to the attached operating procedure**
STATUS: Water flow rate low	**Status: Water flow rate low.** System traffic is displayed in the lower middle of the interface. When the circulating cooling water flow speed in the main unit is lower than 1.6L/min, the system will give an alarm and the alarm prompt will appear in the status bar.	**1. If the flow rate value displayed on the interface is "0 L/min", then:** **1) The device is turned on without a handle attached. The solution is to insert the laser handle.** **2) Signal problem, which refers to the virtual connection between the flow sensor and the control board.** **3) The board is damaged, generally the middle white optocoupler OW1 is virtual connected.** **4) The flow sensor is damaged.** **2. If the interface shows that the flow rate value is lower than 1.6L/min: when the device is shipped from the factory, the water flow rate is between 2.3-2.8L/min. If the actual flow rate rate is lower than 1.6L/min, the cause may be:** **1) The hand is not in contact with the host connector. The CPC water connector is in a semi-contact state and the flow rate is low. Reconnect the laser handpiece at this point.** **2) The air pressure inside the water circulation system is not uniform with the outside air. At this time, the water injection port can be opened by turning on the water injection port joint so that the inside of the circulating water is balanced with the external atmospheric pressure.** Long-term use, individual pumps are aging or damaged. Here you need to check each pump. The troubleshooting method is to first reduce a power supply line of a water pump in order, and compare the actual flow rate. If the flow of a water pump is disconnected and the flow is reduced, it is proved that the disconnected pump is working properly. If the water flow does not get smaller,

		the disconnected pump is damaged. And so on. Check all pumps to find out all the damaged pumps. Replace with a spare pump. This step can be referred to the attachment or click on this link to query the video. 1) The filter is aging and block, replace the filter. The ion filter color can be observed and if it is dark, it needs to be replaced. Before replacing the filter, please use the mobile phone to take a picture of the original filter installation direction, replace the new filter, the direction should be the same as the original. When replacing, pay special attention to the direction of the arrow of the filter bottle. The new filter arrow after replacement should be in the same direction as the arrow of the original filter. Incorrect orientation of the filter can cause the exchange resin in the ion filter to flow out of the entire equipment circulating water system, eventually clogging the treatment handle. This step can be clicked on this link to query the instructional video. 2) Blockage inside the handle or inside the water pipe. Contact the factory or local dealer to replace the handle.
STATUS: Filter and Deionized water need to be changed	**Status: Please replace the cooling water and ion filter.**	1. Observe the ionic meter on the back of the machine at this time, and give priority believe to the reading of the ionic meter. If the reading is greater than 20, that means the water quality is poor. To drain the water, you must add pure water, and then wait for the machine to run for half an hour. During the observation of the meter reading, it should continue to drop until it is below 20, and the alarm disappears. 2. If the ionic meter reading is below 20, the machine still alarm. The solution is as follows: 1) Check if the main control board and the ionic meter head connection are loose and tighten it. 2) Update the program. (The update procedure is actually to remove the alarm settings, this is an emergency means, not recommended)
STATUS: ERROR 02	**Status (Status Bar): Error 2**	1. 1.Is the fan of the control board heat sink working normally. 2. Poor heat dissipation inside the machine or excessive ambient temperature. It can be started after cooling.

10.3 The alarm that will appear after pressing "ready":

Prompt picture	Interpretation	Solution
状态栏: laser is disconnected	The Laser stack disconnection: Laser stack disconnection or poor contact between the laser stack and the wire in the handle.	1. 1. start from the the control board to the laser stack in the handpiece, check if the line is loose. 2. The laser stack inner disconnection and damaged. If there is a spare laser stack, replace it. And contact the manufacturer.

10.4 the alarm appears when you Press the button on the laser handle or step on the foot pedal :

Prompt picture	Interpretation	Solution
STATUS: Operation wrong	**Status: Operation wrong.** The system needs to enter the ready state before the laser can be turned on. In other states, an operation error message will appear when you press the button on the hand or press the foot pedal.	Release the switch and press the "Ready" button on the screen.
STATUS: Please check diode laser or diode driver	**Status: Please check diode laser or diode driver** The system detects that the number of laser bars in the laser handle has been damaged by more than two pieces, or that the output voltage of the switching power supply is too high relative to the load laser or that the control circuit is damaged.	First confirm if it is the laser bar problem: 1. Shut down, restart the device, enter the ready state, observe the light exit of the laser handpiece, observe that how many laser bars still have a weak red light, and how many laser bars are not bright. 2. If the diode laser stack damages more than 3 pieces laser bars, please contact the factory to replace the laser stack. 3. If there is only 1-2 pieces laser bar not bright, you still want to continue using it. You can re-updated the motherboard program. Please refer to the update procedure. Or adjust the output voltage of the switching power supply, about 2-4V, to reduce the output voltage of 2v for each bar damaged. This is an emergency

		operation, and it is recommended to replace the new laser with long-term use.